

Exercise Set 3

Problem 1

A research project investigates the relationship between smoking habits and gender. The probability of smoking is described by the following equation:

$$P(\text{smoke} = 1) = \beta_0 + \beta_1 \text{female} + \beta_2 \text{smkban} + \beta_3 \text{age} + u. \quad (1)$$

The variables are defined as follows:

smoke = 1 if the person is a smoker, 0 otherwise;
female = 1 if the person is a female, 0 otherwise;
smkban = 1 if there is a smoking ban at the place of residence;
age = age in years.

u denotes an i.i.d. error term.

- 1.1 One option to estimate the probability $P(\text{smoke} = 1)$ is the linear probability model. What are the advantages and disadvantages of using a linear probability model?
- 1.2 To achieve a better fit of the model to the data, one can estimate the same relationship using a logit model. Set up the likelihood function and the log-likelihood function for the logit model.
- 1.3 Estimating the logit model with Stata yields the results reported in Table 1. Interpret the coefficient of *smkban*. Why would we expect a negative coefficient?

Table 1: regression results: Logit model

Logistic regression	Number of obs	=	10,000
	LR chi2(3)	=	90.82
	Prob > chi2	=	0.0000
Log likelihood = -5491.7578	Pseudo R2	=	0.0082

smoke	Coef.	Std. Err.	z P> z [95% Conf. Interval]

female	-.0987092	.0473998	-2.08 0.037 -.1916112 -.0058072
smkban	-.3983811	.0475606	-8.38 0.000 -.4915982 -.305164
age	-.0059029	.0019526	-3.02 0.003 -.00973 -.0020759
_cons	-.6256074	.0843635	-7.42 0.000 -.7909569 -.4602579

- 1.4 Calculate the probability that a 30-year-old man smokes despite an existing smoking ban. Hint:

$$P(y_i = 1) = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})}.$$

- 1.5 Calculate and interpret the marginal effect of *age* for a 30-year-old man who smokes despite an existing smoking ban. Hint:

$$\frac{\partial P(y_i = 1)}{\partial x_{ik}} = \frac{\beta_k \exp(\mathbf{x}_i' \boldsymbol{\beta})}{(1 + \exp(\mathbf{x}_i' \boldsymbol{\beta}))^2}.$$

Problem 2

Let us consider the following simple model of the labor supply of married women:

$$lfp = \beta_0 + \beta_1 k5 + \beta_2 k618 + \beta_3 age + \beta_4 wc + \beta_5 hc + \beta_6 lwg + \beta_7 inc + e_i \quad (2)$$

where:

- lfp = 1 if the woman is in the paid labour force, 0 otherwise,
- $k5$ = number of children aged 5 and younger,
- $k618$ = number of children aged 6 to 18,
- age = woman's age in years,
- wc = 1 if woman attended college, 0 otherwise,
- hc = 1 if partner attended college, 0 otherwise,
- lwg = log of woman's estimated wage rate,
- inc = family income excluding woman's wages, 1000 dollars.

- 2.1 Load the data (`Mroz{carData}`). Tabulate the descriptive statistics for the variables listed above. Compare the descriptive statistics of women in the labor force to those not participating in the labor force.
- 2.2 Estimate a logit model for lfp using all of the above indicated variables and interpret the results.
- 2.3 Let us have a closer look at inc and $k5$. Compute the average marginal effect and the marginal effect at the average.
- 2.4 Perform a likelihood ratio test for the joint significance of the variables lwg and inc .
- 2.5 Fit the model again using a Bayesian approach.

Problem 3

Use the dataset *loanapp* of the Wooldridge-package, a dataset containing 1989 observations on 59 variables. The binary variable to be explained is *approve*, which is equal to one if a mortgage loan to an individual was approved. Consider the following model:

$$approve = \beta_0 + \beta_1 white + \beta_2 male + \beta_3 married + \beta_4 unem + e \quad (3)$$

- $approve$ = 1 if loan is approved, 0 otherwise,
- $white$ = 1 if applicant is white, 0 otherwise
- $male$ = 1 if applicant is male, 0 otherwise
- $married$ = 1 if applicant is married, 0 otherwise
- $unem$ = unemployment rate by industry,

- 3.1 Estimate a linear probability model, a logit model and a probit model. Summarize your results in a table.
- 3.2 Compute the average marginal effects as well as the marginal effects at the average.

References

- Mroz, Thomas A. (1987). *The Sensitivity of an Empirical Model of Married Women's Hours of Work to Economic and Statistical Assumptions*. *Econometrica*, 55, 765-799.
- Wooldridge, Jeffrey M. (2013). *Introductory Econometrics. A modern Approach*. South-Western, Cengage Learning.